

website, is that Moloch augments the existing security infrastructure by storing and indexing network traffic in standard PCAP format, while also providing fast indexed access. Moloch is not meant to replace intrusion detection systems (IDS); instead, it provides more visibility. Moloch is built with an intuitive UI/UX which reduces the analysis time for suspected incidents.

The core features work towards reproducibility, an intuitive Web interface, scalability, and the provision of application programmable interfaces to allow JSON and PCAP data to be downloaded directly.

GRR Rapid Response: Google's GRR Rapid Response expands its remote forensics capabilities in order to be an incident-response framework. Its goal is to support forensics and investigation in a scalable manner, and to allow analysts to rapidly triage attacks and perform analysis remotely, within a short time frame. The software follows a client-server architecture, with the client deployed on systems to be investigated, and the server comprising several components with an API endpoint allowing the scheduling of actions on clients.

The toolkit offers a broad range of functionalities using a collection of libraries including Yara and the Sleuth Kit. It is built to scale and to maintain the granularity of events, even as it offers a high-level overview of an incident. It follows an asynchronous design allowing for automated scheduling, adding to its already-powerful arsenal of functionalities across enterprise use cases.

Security Onion: Kali Linux is a distribution that is typically used for aggressive penetration testing, but when it comes to network security, the Security Onion distro is what many cyber security researchers are using. It has been

growing slowly but steadily since its release in 2014 and holds a well-deserved place on this list.

Security Onion comprises a host of tools that come bundled within, saving a significant amount of setup time and effort. The website claims it performs intrusion detection, enterprise security monitoring and log management, and that it includes Elasticsearch, Logstash, Kibana, Snort, Suricata, Bro, Wazuh, Sguil, Squert, CyberChef, NetworkMiner, and many other security tools. The easy-to-use setup wizard allows you to build an army of distributed sensors for your enterprise in minutes.

The old guard

These are the good old tools that have been tried and tested by generations of security researchers and enthusiasts, and still remain popular in the field for a variety of audits and vulnerability scans.

OpenVAS: A comprehensive toolkit spun off from the erstwhile, popular Nessus software (which turned to a proprietary licence), OpenVAS is a tool offering over 50,000 vulnerability scans as documented by Greenbone Networks, since 2007. As per the official website, its capabilities include unauthenticated testing, authenticated testing, various high level and low level Internet and industrial protocols, performance tuning for large scale scans, and a powerful internal programming language to implement any type of vulnerability test.

NMAP: A name that has become synonymous with port scanning, NMAP is a tool that offers a deep set of functionalities targeted at helping users map open ports and running

services better, along with the network and operating system fingerprinting at industrial scale. It offers powerful NSE scripts that can detect vulnerabilities and misconfigurations in computer networks.

OSSMAP: The website for OSSEC lists its purpose as watching, alerting and being everywhere. A popular network and host monitoring system, OSSMAP performs log analysis, rootkit detection, real-time alerting, integrity checking, Windows registry monitoring, and active response.

WireShark: WireShark is to network protocol analysis what NMAP is to port scanning. It lets you see what's happening on your network at a microscopic level and is the go-to solution across many commercial and non-profit enterprises, government agencies and educational institutions.

The Metasploit Framework: Scanning, auditing, and exploits — the Metasploit Framework serves as the primary offensive tool for a number of penetration testers in the field. It does more than simply verify vulnerabilities and manage security assessments, given its wide range of functionalities targeted at security awareness, enabling defenders to stay a step ahead of the game.

The cyber security world is incredibly complex, volatile and delicate, constantly under DDoS attacks by an assortment of entities around the world. However, amidst all these new entrants into the world of NMAP, Kali Linux, OpenVAS and WireShark, we are entering an exciting era in which even more data vigilance is required. **END** 

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